

Assignment - 2 & 3

1. Version Control helps track changes in code, prevents data loss, and allows multiple developers to work on the same project without conflicts.
2. A **Version Control System (VCS)** is a tool that records changes to files over time, allowing users to manage, track, and restore previous versions.
3. **Git** is a distributed version control system used to track code changes, collaborate with others, and manage software projects efficiently.
4. **Git** is a version control tool used to manage source code locally. **GitHub** is an online platform that hosts Git repositories and enables collaboration, sharing, and project management.
5. A **Repository (Repo)** is a storage location where a project's files, code, and version history are maintained using Git.
6. A **commit** is a saved snapshot of the changes made to a project. Each commit records the changes along with a unique ID and a message.
7. The **Staging Area** is used to prepare and review selected changes before saving them as a commit.
8. **Working Directory:** The folder where you create and modify project files.
Staging Area: A temporary area where selected changes are kept before committing.
Local Repository: The Git repository on your computer that stores the project's commit history.
Remote Repository: The online repository (such as GitHub) used to store, share, and collaborate on the project with others.
9. **a) git init**-Initializes a new Git repository in the current folder.
b) git status-Displays the current status of the repository, including modified and staged files.
c) git add .-Adds all modified and new files to the Staging Area.
d) git commit -m "message"-Saves the staged changes to the local repository with a descriptive commit message.
e) git push-Uploads the committed changes from the local repository to the remote repository (GitHub).
10.
 - Clone the repository using git clone.
 - Make the required changes in the project files.
 - Check the changes using git status.
 - Add the changes to the Staging Area using git add ..

- Commit the changes using `git commit -m "message"`.
 - Push the committed changes to the remote repository using `git push`.
11. A Git commit stores the changes made to the project, a unique commit ID, the author's details, the date and time, and the commit message.
 12. Git is a Distributed Version Control System because every developer has a complete copy of the repository and its history on their local computer.
 13. GitHub provides online repository hosting, easy collaboration, code backup, version history, and project sharing with team members.
 14. **Push:** Uploads local commits to the remote repository
Pull: Downloads the latest changes from the remote repository to the local repository.
 15. Git allows multiple developers to work on the same project, track changes, merge code, resolve conflicts, and maintain version history efficiently.

Part-B

Task-1

```
PS C:\Users\Roli Johri> git --version
git version 2.54.0.windows.1
PS C:\Users\Roli Johri> git config --global user.name "Roli Johari"
PS C:\Users\Roli Johri> git config --global user.name
Roli Johari
PS C:\Users\Roli Johri> git config --global user.email "rolijohari48@gmail.com"
PS C:\Users\Roli Johri> git config --global user.email
rolijohari48@gmail.com
PS C:\Users\Roli Johri>
```

Task-2

```
PS C:\Users\Roli Johri\Documents> mkdir git-assignment

Directory: C:\Users\Roli Johri\Documents

Mode                LastWriteTime         Length Name
----                -
d-----          01-07-2026   20:41             git-assignment

PS C:\Users\Roli Johri\Documents> cd git-assignment
```

```
PS C:\Users\Roli Johri\Documents\git-assignment> git init
Initialized empty Git repository in C:/Users/Roli Johri/Documents/git-assignment/.git/
PS C:\Users\Roli Johri\Documents\git-assignment> git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        index.html

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\Roli Johri\Documents\git-assignment>
```

Task-3

```
PS C:\Users\Roli Johri\Documents\git-assignment> git add .
PS C:\Users\Roli Johri\Documents\git-assignment> git commit -m "Initial Commit"
[master (root-commit) 2854b60] Initial Commit
1 file changed, 11 insertions(+)
create mode 100644 index.html
PS C:\Users\Roli Johri\Documents\git-assignment> git log
commit 2854b6065915b7846c27929ed35761e7f009cbbc (HEAD -> master)
Author: Roli Johari <rolijohari48@gmail.com>
Date:   Wed Jul 1 20:52:14 2026 +0530

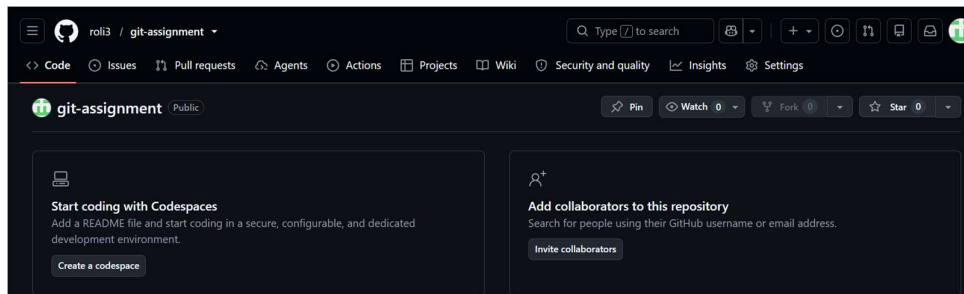
    Initial Commit
PS C:\Users\Roli Johri\Documents\git-assignment> git status
On branch master
nothing to commit, working tree clean
```

Task -4

```
PS C:\Users\Roli Johri\Documents\git-assignment> git status
On branch master
nothing to commit, working tree clean
PS C:\Users\Roli Johri\Documents\git-assignment> git add .
PS C:\Users\Roli Johri\Documents\git-assignment> git commit -m "Added Student Name"
On branch master
nothing to commit, working tree clean
PS C:\Users\Roli Johri\Documents\git-assignment> git log
commit 2854b6065915b7846c27929ed35761e7f009cbbc (HEAD -> master)
Author: Roli Johari <rolijohari48@gmail.com>
Date:   Wed Jul 1 20:52:14 2026 +0530

    Initial Commit
```

Task -5

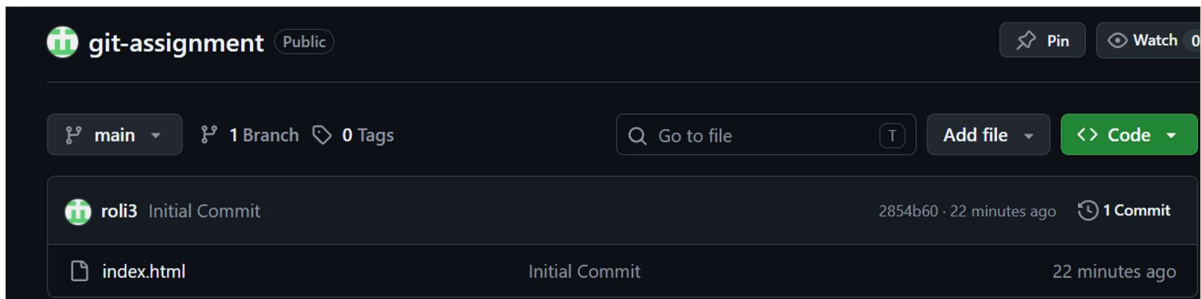


Task -6

```
Initial Commit
PS C:\Users\Roli Johri\Documents\git-assignment> git remote add origin https://github.com/roli3/git-assignment.git
PS C:\Users\Roli Johri\Documents\git-assignment> git remote -v
origin https://github.com/roli3/git-assignment.git (fetch)
origin https://github.com/roli3/git-assignment.git (push)
```

Task - 7

```
PS C:\Users\Roli Johri\Documents\git-assignment> git branch -M main
PS C:\Users\Roli Johri\Documents\git-assignment> git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 387 bytes | 387.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/roli3/git-assignment.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```



Task-8

